Neural Computation -- 2022

81 article(s), all issues

1. Neuromorphic Engineering: In Memory of Misha Mahowald

Carver Mead

https://doi.org/10.1162/neco_a_01553 | pp. 1-41

We review the coevolution of hardware and software dedicated to neuromorphic systems. From modest beginnings, these disciplines have become central to the larger field of computation. In the process, their biological foundations become more relevant, and their realizations increasingly overlap. We identify opportunities for significant steps forward in both the near and more distant future.

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2. On the Explainability of Graph Convolutional Network With GCN Tangent Kernel

Xianchen Zhou, Hongxia Wang

https://doi.org/10.1162/neco_a_01548 | pp. 1-26

Graph convolutional network (GCN) is a powerful deep model in dealing with graph data. However, the explainability of GCN remains a difficult problem since the training behaviors for graph neural networks are hard to describe. In this work, we show that for GCN with wide hidden feature dimension, the output for semisupervised problem can be described by a simple differential equation. In addition, the dynamics of the behavior of output is decided by the graph convolutional neural tangent kernel (GCNTK), which is stable when the width of hidden feature tends to be infinite. And the solution of node classification can be explained directly by the differential equation for a semisupervised problem. The experiments on some toy models speak to the consistency of the GCNTK model and GCN.

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3. Visuomotor Mismatch Responses as a Hallmark of Explaining Away in Causal Inference

Fabian A. Mikulasch, Lucas Rudelt, Viola Priesemann

https://doi.org/10.1162/neco_a_01546 | pp. 27-37

How are visuomotor mismatch responses in primary visual cortex embedded into cortical processing? We here show that mismatch responses can be understood as the result of a cooperation of motor and visual areas to jointly explain optic flow. This cooperation requires that optic flow is not explained redundantly by both areas, meaning that optic flow inputs to V1 that are predictable from motor neurons should be canceled (i.e., explained away). As a result, neurons in V1 represent only external causes of optic flow, which could allow the animal to easily detect movements that are independent of its own locomotion. We implement the proposed model in a spiking neural network, where coding errors are computed in dendrites and synaptic weights are learned with voltage-dependent plasticity rules. We find that both positive and negative mismatch responses arise, providing an alternative to the prevailing idea that visuomotor mismatch responses are linked to dedicated neurons for error computation. These results also provide a new perspective on several other recent observations of cross-modal neural interactions in cortex.

4. Progressive Interpretation Synthesis: Interpreting Task Solving by Quantifying Previously Used and Unused Information

Zhengqi He, Taro Toyoizumi

https://doi.org/10.1162/neco_a_01542 | pp. 38-57

A deep neural network is a good task solver, but it is difficult to make sense of its operation. People have different ideas about how to interpret its operation. We look at this problem from a new perspective where the interpretation of task solving is synthesized by quantifying how much and what previously unused information is exploited in addition to the information used to solve previous tasks. First, after learning several tasks, the network acquires several information partitions related to each task. We propose that the network then learns the minimal information partition that supplements previously learned information partitions to more accurately represent the input. This extra partition is associated with unconceptualized information that has not been used in previous tasks. We manage to identify what unconceptualized information is used and quantify the amount. To interpret how the network solves a new task, we quantify as meta-information how much information from each partition is extracted. We implement this framework with the variational information bottleneck technique. We test the framework with the MNIST and the CLEVR data set. The framework is shown to be able to compose information partitions and synthesize experience-dependent interpretation in the form of meta-information. This system progressively improves the resolution of interpretation upon new experience by converting a part of the unconceptualized information partition to a task-related partition. It can also provide a visual interpretation by imaging what is the part of previously unconceptualized information that is needed to solve a new task.

5. Learning With Proper Partial Labels

Zhenguo Wu, Jiaqi Lv, Masashi Sugiyama

https://doi.org/10.1162/neco_a_01554 | pp. 58-81

Partial-label learning is a kind of weakly supervised learning with inexact labels, where for each training example, we are given a set of candidate labels instead of only one true label. Recently, various approaches on partial-label learning have been proposed under different generation models of candidate label sets. However, these methods require relatively strong distributional assumptions on the generation models. When the assumptions do not hold, the performance of the methods is not guaranteed theoretically. In this letter, we propose the notion of properness on partial labels. We show that this proper partial-label learning framework requires a weaker distributional assumption and includes many previous partial-label learning settings as special cases. We then derive a unified unbiased estimator of the classification risk. We prove that our estimator is risk consistent, and we also establish an estimation error bound. Finally, we validate the effectiveness of our algorithm through experiments.

6. Gaussian Process Koopman Mode Decomposition

Takahiro Kawashima, Hideitsu Hino

https://doi.org/10.1162/neco_a_01555 | pp. 82-103

We propose a nonlinear probabilistic generative model of Koopman mode decomposition based on an unsupervised gaussian process. Existing data-driven methods for Koopman mode decomposition have focused on estimating the quantities specified by Koopman mode decomposition: eigenvalues, eigenfunctions, and modes. Our model enables the simultaneous estimation of these quantities and latent variables governed by an unknown dynamical system. Furthermore, we introduce an efficient strategy to estimate the parameters of our model by low-rank approximations of covariance matrices. Applying the proposed model to both synthetic data and a real-world epidemiological data set, we show that various analyses are available using the estimated parameters.

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7. Optimal Quadratic Binding for Relational Reasoning in Vector Symbolic Neural Architectures

Naoki Hiratani, Haim Sompolinsky

https://doi.org/10.1162/neco_a_01558 | pp. 105-155

Binding operation is fundamental to many cognitive processes, such as cognitive map formation, relational reasoning, and language comprehension. In these processes, two different modalities, such as location and objects, events and their contextual cues, and words and their roles, need to be bound together, but little is known about the underlying neural mechanisms. Previous work has introduced a binding model based on quadratic functions of bound pairs, followed by vector summation of multiple pairs. Based on this framework, we address the following questions: Which classes of quadratic matrices are optimal for decoding relational structures? And what is the resultant accuracy? We introduce a new class of binding matrices based on a matrix representation of octonion algebra, an eight-dimensional extension of complex numbers. We show that these matrices enable a more accurate unbinding than previously known methods when a small number of pairs are present. Moreover, numerical optimization of a binding operator converges to this octonion binding. We also show that when there are a large number of bound pairs, however, a random quadratic binding performs, as well as the octonion and previously proposed binding methods. This study thus provides new insight into potential neural mechanisms of binding operations in the brain.

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8. The Tensor Brain: A Unified Theory of Perception, Memory, and Semantic Decoding

Volker Tresp, Sahand Sharifzadeh, Hang Li, Dario Konopatzki, Yunpu Ma

https://doi.org/10.1162/neco_a_01552 | pp. 156-227

We present a unified computational theory of an agent's perception and memory. In our model, both perception and memory are realized by different operational modes of the oscillating interactions between a symbolic index layer and a subsymbolic representation layer. The two layers form a bilayer tensor network (BTN). The index layer encodes indices for concepts, predicates, and episodic instances. The representation layer broadcasts information and reflects the cognitive brain state; it is our model of what authors have called the "mental canvas" or the "global workspace." As a bridge between perceptual input and the index layer, the representation layer enables the grounding of indices by their subsymbolic embeddings, which are implemented as connection weights linking both layers. The propagation of activation to earlier perceptual processing layers in the brain can lead to embodiments of indices. Perception and memories first create subsymbolic representations, which are subsequently decoded semantically to

produce sequences of activated indices that form symbolic triple statements. The brain is a sampling engine: only activated indices are communicated to the remaining parts of the brain. Triple statements are dynamically embedded in the representation layer and embodied in earlier processing layers: the brain speaks to itself. Although memory appears to be about the past, its main purpose is to support the agent in the present and the future. Recent episodic memory provides the agent with a sense of the here and now. Remote episodic memory retrieves relevant past experiences to provide information about possible future scenarios. This aids the agent in decision making. "Future" episodic memory, based on expected future events, guides planning and action. Semantic memory retrieves specific information, which is not delivered by current perception, and defines priors for future observations. We argue that it is important for the agent to encode individual entities, not just classes and attributes. Perception is learning: episodic memories are constantly being formed, and we demonstrate that a form of self-supervised learning can acquire new concepts and refine existing ones. We test our model on a standard benchmark data set, which we expanded to contain richer representations for attributes, classes, and individuals. Our key hypothesis is that obtaining a better understanding of perception and memory is a crucial prerequisite to comprehending human-level intelligence.

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9. Dynamic Consolidation for Continual Learning

Hang Li, Chen Ma, Xi Chen, Xue Liu

https://doi.org/10.1162/neco_a_01560 | pp. 228-248

Training deep learning models from a stream of nonstationary data is a critical problem to be solved to achieve general artificial intelligence. As a promising solution, the continual learning (CL) technique aims to build intelligent systems that have the plasticity to learn from new information without forgetting the previously obtained knowledge. Unfortunately, existing CL methods face two nontrivial limitations. First, when updating a model with new data, existing CL methods usually constrain the model parameters within the vicinity of the parameters optimized for old data, limiting the exploration ability of the model; second, the important strength of each parameter (used to consolidate the previously learned knowledge) is fixed and thus is suboptimal for the dynamic parameter updates. To address these limitations, we first relax the vicinity constraints with a global definition of the important strength, which allows us to explore the full parameter space. Specifically, we define the important strength as the sensitivity of the global loss function to the model parameters. Moreover, we propose adjusting the important strength adaptively to align it with the dynamic parameter updates. Through extensive experiments on popular data sets, we demonstrate that our proposed method outperforms the strong baselines by up to 24% in terms of average accuracy.

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10. Identifying and Localizing Multiple Objects Using Artificial Ventral and Dorsal Cortical Visual Pathways

Zhixian Han, Anne B. Sereno

https://doi.org/10.1162/neco_a_01559 | pp. 249-275

In our previous study (Han & Sereno, 2022a), we found that two artificial cortical visual pathways trained for either identity or space actively retain information about both identity and space independently and differently. We also found that this independently and differently retained information about identity and space in two separate pathways may be necessary to accurately and optimally recognize and localize

objects. One limitation of our previous study was that there was only one object in each visual image, whereas in reality, there may be multiple objects in a scene. In this study, we find we are able to generalize our findings to object recognition and localization tasks where multiple objects are present in each visual image. We constrain the binding problem by training the identity network pathway to report the identities of objects in a given order according to the relative spatial relationships between the objects, given that most visual cortical areas including high-level ventral steam areas retain spatial information. Under these conditions, we find that the artificial neural networks with two pathways for identity and space have better performance in multiple-objects recognition and localization tasks (higher average testing accuracy, lower testing accuracy variance, less training time) than the artificial neural networks with a single pathway. We also find that the required number of training samples and the required training time increase quickly, and potentially exponentially, when the number of objects in each image increases, and we suggest that binding information from multiple objects simultaneously within any network (cortical area) induces conflict or competition and may be part of the reason why our brain has limited attentional and visual working memory capacities.

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11. Feelings Are the Source of Consciousness

António R. Damásio, Hanna Damásio

https://doi.org/10.1162/neco_a_01521 | pp. 277-286

In this view, we address the problem of consciousness, and although we focus on its human presentation, we note that the phenomenon is present in numerous nonhuman species and use findings from a variety of animal studies to explain our hypothesis for how consciousness is made. Consciousness occurs when mind contents, such as perceptions and thoughts, are spontaneously identified as belonging to a specific organism/owner. Conscious minds are said to have a self that experiences mental events. We hypothesize that the automatic identification that associates minds and organisms is provided by a continuous flow of homeostatic feelings. Those feelings arise from the uninterrupted process of life regulation and correspond to both salient physiological fluctuations such as hunger, pain, well-being, or malaise, as well as to states closer to metabolic equilibrium and best described as feelings of life/existence, such as breathing or body temperature. We also hypothesize that homeostatic feelings were the inaugural phenomena of consciousness in biological evolution and venture that they were selected because the information they provided regarding the current state of life regulation conferred extraordinary advantages to the organisms so endowed. The "knowledge" carried by conscious homeostatic feelings provided "overt" guidance for life regulation, an advance over the covert regulation present in nonconscious organisms. Finally, we outline a mechanism for the generation of feelings based on a two-way interaction between interoceptive components of the nervous system and a particular set of nonneural components of the organism's interior, namely, viscera and circulating chemical molecules involved in their operations. Feelings emerge from this interaction as continuous and hybrid phenomena, related simultaneously to two series of events. The first is best described by the terms neural/representational/and mental and the second by the terms nonneural/visceral/and chemical. We note that this account offers a solution for the mind-body problem: homeostatic feelings constitute the "mental" version of bodily processes.

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12. Understanding and Applying Deep Learning

Richard P. Lippmann

https://doi.org/10.1162/neco_a_01518 | pp. 287-308

The past 10 years have witnessed an explosion in deep learning neural network model development. The most common perceptual models with vision, speech, and text inputs are not general-purpose AI systems but tools. They automatically extract clues from inputs and compute probabilities of class labels. Successful applications require representative training data, an understanding of the limitations and capabilities of deep learning, and careful attention to a complex development process. The goal of this view is to foster an intuitive understanding of convolutional network deep learning models and how to use them with the goal of engaging a wider creative community. A focus is to make it possible for experts in areas such as health, education, poverty, and agriculture to understand the process of deep learning model development so they can help transition effective solutions to practice.

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13. Toward a Biomimetic Neural Circuit Model of Sensory-Motor Processing

Stephen G. Lisberger

https://doi.org/10.1162/neco_a_01516 | pp. 384-412

Computational models have been a mainstay of research on smooth pursuit eye movements in monkeys. Pursuit is a sensory-motor system that is driven by the visual motion of small targets. It creates a smooth eye movement that accelerates up to target speed and tracks the moving target essentially perfectly. In this review of my laboratory's research, I trace the development of computational models of pursuit eye movements from the early control-theory models to the most recent neural circuit models. I outline a combined experimental and computational plan to move the models to the next level. Finally, I explain why research on nonhuman primates is so critical to the development of the neural circuit models I think we need.

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14. How to Represent Part-Whole Hierarchies in a Neural Network

Geoffrey E. Hinton

https://doi.org/10.1162/neco_a_01557 | pp. 413-452

This article does not describe a working system. Instead, it presents a single idea about representation that allows advances made by several different groups to be combined into an imaginary system called GLOM.1 The advances include transformers, neural fields, contrastive representation learning, distillation, and capsules. GLOM answers the question: How can a neural network with a fixed architecture parse an image into a part-whole hierarchy that has a different structure for each image? The idea is simply to use islands of identical vectors to represent the nodes in the parse tree. If GLOM can be made to work, it should significantly improve the interpretability of the representations produced by transformer-like systems when applied to vision or language.

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15. Toward Network Intelligence

Alex Pentland

https://doi.org/10.1162/neco_a_01536 | pp. 525-535

This article proposes a conceptual framework to guide research in neural computation by relating it to mathematical progress in other fields and to examples illustrative of biological networks. The goal is to provide insight into how biological networks, and possibly large artificial networks such as foundation models, transition from analog computation to an analog approximation of symbolic computation. From the mathematical perspective, I focus on the development of consistent symbolic representations and optimal policies for action selection within network settings. From the biological perspective, I give examples of human and animal social network behavior that may be described using these mathematical models.

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16. Single Circuit in V1 Capable of Switching Contexts During Movement Using an Inhibitory Population as a Switch

Doris Voina, Stefano Recanatesi, Brian Hu, Eric Shea■Brown, ■tefan Mihala■

https://doi.org/10.1162/neco_a_01472 | pp. 541-594

As animals adapt to their environments, their brains are tasked with processing stimuli in different sensory contexts. Whether these computations are context dependent or independent, they are all implemented in the same neural tissue. A crucial question is what neural architectures can respond flexibly to a range of stimulus conditions and switch between them. This is a particular case of flexible architecture that permits multiple related computations within a single circuit. Here, we address this question in the specific case of the visual system circuitry, focusing on context integration, defined as the integration of feedforward and surround information across visual space. We show that a biologically inspired microcircuit with multiple inhibitory cell types can switch between visual processing of the static context and the moving context. In our model, the VIP population acts as the switch and modulates the visual circuit through a disinhibitory motif. Moreover, the VIP population is efficient, requiring only a relatively small number of neurons to switch contexts. This circuit eliminates noise in videos by using appropriate lateral connections for contextual spatiotemporal surround modulation, having superior denoising performance compared to circuits where only one context is learned. Our findings shed light on a minimally complex architecture that is capable of switching between two naturalistic contexts using few switching units.

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17. Manifold Alignment Aware Ants: A Markovian Process for Manifold Extraction

Mohammad Mohammadi, Peter Ti■o, Kerstin Bunte

https://doi.org/10.1162/neco_a_01478 | pp. 595-641

The presence of manifolds is a common assumption in many applications, including astronomy and computer vision. For instance, in astronomy, low-dimensional stellar structures, such as streams, shells, and globular clusters, can be found in the neighborhood of big galaxies such as the Milky Way. Since these structures are often buried in very large data sets, an algorithm, which can not only recover the manifold but also remove the background noise (or outliers), is highly desirable. While other works try to recover manifolds either by pushing all points toward manifolds or by downsampling from dense regions, aiming to solve one of the problems, they generally fail to suppress the noise on manifolds and remove background noise simultaneously. Inspired by the collective behavior of biological ants in food-seeking process, we propose a new algorithm that employs several random walkers equipped with a local alignment measure to detect and denoise manifolds. During the walking process, the agents release pheromone on data points,

which reinforces future movements. Over time the pheromone concentrates on the manifolds, while it fades in the background noise due to an evaporation procedure. We use the Markov chain (MC) framework to provide a theoretical analysis of the convergence of the algorithm and its performance. Moreover, an empirical analysis, based on synthetic and real-world data sets, is provided to demonstrate its applicability in different areas, such as improving the performance of t-distributed stochastic neighbor embedding (t-SNE) and spectral clustering using the underlying MC formulas, recovering astronomical low-dimensional structures, and improving the performance of the fast Parzen window density estimator.

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18. Predicting the Future With a Scale-Invariant Temporal Memory for the Past

Wei Zhong Goh, Varun Ursekar, Marc W. Howard

https://doi.org/10.1162/neco_a_01475 | pp. 642-685

In recent years, it has become clear that the brain maintains a temporal memory of recent events stretching far into the past. This letter presents a neurally inspired algorithm to use a scale-invariant temporal representation of the past to predict a scale-invariant future. The result is a scale-invariant estimate of future events as a function of the time at which they are expected to occur. The algorithm is time-local, with credit assigned to the present event by observing how it affects the prediction of the future. To illustrate the potential utility of this approach, we test the model on simultaneous renewal processes with different timescales. The algorithm scales well on these problems despite the fact that the number of states needed to describe them as a Markov process grows exponentially.

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19. Active Role of Self-Sustained Neural Activity on Sensory Input Processing: A Minimal Theoretical Model

Bruno André Santos, Rogério Martins Gomes, Xabier E. Barandiaran, Phil Husbands

https://doi.org/10.1162/neco_a_01471 | pp. 686-715

A growing body of work has demonstrated the importance of ongoing oscillatory neural activity in sensory processing and the generation of sensorimotor behaviors. It has been shown, for several different brain areas, that sensory-evoked neural oscillations are generated from the modulation by sensory inputs of inherent self-sustained neural activity (SSA). This letter contributes to that strand of research by introducing a methodology to investigate how much of the sensory-evoked oscillatory activity is generated by SSA and how much is generated by sensory inputs within the context of sensorimotor behavior in a computational model. We develop an abstract model consisting of a network of three Kuramoto oscillators controlling the behavior of a simulated agent performing a categorical perception task. The effects of sensory inputs and SSAs on sensory-evoked oscillations are quantified by the cross product of velocity vectors in the phase space of the network under different conditions (disconnected without input, connected without input, and connected with input). We found that while the agent is carrying out the task, sensory-evoked activity is predominantly generated by SSA (93.10%) with much less influence from sensory inputs (6.90%). Furthermore, the influence of sensory inputs can be reduced by 10.4% (from 6.90% to 6.18%) with a decay in the agent's performance of only 2%. A dynamical analysis shows how sensory-evoked oscillations are generated from a dynamic coupling between the level of sensitivity of the network and the intensity of the input signals. This work may suggest interesting directions for neurophysiological experiments investigating

how self-sustained neural activity influences sensory input processing, and ultimately affects behavior.

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20. TARA: Training and Representation Alteration for AI Fairness and Domain Generalization

William Paul, Armin Hadži■, Neil Joshi, Fady Alajaji, Phil Burlina

https://doi.org/10.1162/neco_a_01468 | pp. 716-753

We propose a novel method for enforcing AI fairness with respect to protected or sensitive factors. This method uses a dual strategy performing training and representation alteration (TARA) for the mitigation of prominent causes of AI bias. It includes the use of representation learning alteration via adversarial independence to suppress the bias-inducing dependence of the data representation from protected factors and training set alteration via intelligent augmentation to address bias-causing data imbalance by using generative models that allow the fine control of sensitive factors related to underrepresented populations via domain adaptation and latent space manipulation. When testing our methods on image analytics, experiments demonstrate that TARA significantly or fully debiases baseline models while outperforming competing debiasing methods that have the same amount of information-for example, with (% overall accuracy, % accuracy gap) = (78.8, 0.5) versus the baseline method's score of (71.8, 10.5) for Eye-PACS, and (73.7, 11.8) versus (69.1, 21.7) for CelebA. Furthermore, recognizing certain limitations in current metrics used for assessing debiasing performance, we propose novel conjunctive debiasing metrics. Our experiments also demonstrate the ability of these novel metrics in assessing the Pareto efficiency of the proposed methods.

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21. Understanding Memories of the Past in the Context of Different Complex Neural Network Architectures

C. Bohm, Douglas Kirkpatrick, Arend Hintze

https://doi.org/10.1162/neco_a_01469 | pp. 754-780

Deep learning (primarily using backpropagation) and neuroevolution are the preeminent methods of optimizing artificial neural networks. However, they often create black boxes that are as hard to understand as the natural brains they seek to mimic. Previous work has identified an information-theoretic tool, referred to as R, which allows us to quantify and identify mental representations in artificial cognitive systems. The use of such measures has allowed us to make previous black boxes more transparent. Here we extend R to not only identify where complex computational systems store memory about their environment but also to differentiate between different time points in the past. We show how this extended measure can identify the location of memory related to past experiences in neural networks optimized by deep learning as well as a genetic algorithm.

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22. Active Classification With Uncertainty Comparison Queries

Zhenghang Cui, Issei Sato

https://doi.org/10.1162/neco_a_01473 | pp. 781-803

Noisy pairwise comparison feedback has been incorporated to improve the overall query complexity of interactively learning binary classifiers. The positivity comparison oracle is extensively used to provide feedback on which is more likely to be positive in a pair of data points. Because it is impossible to determine accurate labels using this oracle alone without knowing the classification threshold, existing methods still rely on the traditional explicit labeling oracle, which explicitly answers the label given a data point. The current method conducts sorting on all data points and uses explicit labeling oracle to find the classification threshold. However, it has two drawbacks: (1) it needs unnecessary sorting for label inference and (2) it naively adapts quick sort to noisy feedback. In order to avoid these inefficiencies and acquire information of the classification threshold at the same time, we propose a new pairwise comparison oracle concerning uncertainties. This oracle answers which one has higher uncertainty given a pair of data points. We then propose an efficient adaptive labeling algorithm to take advantage of the proposed oracle. In addition, we address the situation where the labeling budget is insufficient compared to the data set size. Furthermore, we confirm the feasibility of the proposed oracle and the performance of the proposed algorithm theoretically and empirically.

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23. Dynamical Mechanism of Sampling-Based Probabilistic Inference Under Probabilistic Population Codes

Kohei Ichikawa, Asaki Kataoka

https://doi.org/10.1162/neco_a_01477 | pp. 804-827

Animals make efficient probabilistic inferences based on uncertain and noisy information from the outside environment. It is known that probabilistic population codes, which have been proposed as a neural basis for encoding probability distributions, allow general neural networks (NNs) to perform near-optimal point estimation. However, the mechanism of sampling-based probabilistic inference has not been clarified. In this study, we trained two types of artificial NNs, feedforward NN (FFNN) and recurrent NN (RNN), to perform sampling-based probabilistic inference. Then we analyzed and compared their mechanisms of sampling. We found that sampling in RNN was performed by a mechanism that efficiently uses the properties of dynamical systems, unlike FFNN. In addition, we found that sampling in RNNs acted as an inductive bias, enabling a more accurate estimation than in maximum a posteriori estimation. These results provide important arguments for discussing the relationship between dynamical systems and information processing in NNs.

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24. Bayesian Brains and the Rényi Divergence

Noor Sajid, Francesco Faccio, Lancelot Da Costa, Thomas Parr, Jürgen Schmidhuber, Karl Friston

https://doi.org/10.1162/neco_a_01484 | pp. 829-855

Under the Bayesian brain hypothesis, behavioral variations can be attributed to different priors over generative model parameters. This provides a formal explanation for why individuals exhibit inconsistent behavioral preferences when confronted with similar choices. For example, greedy preferences are a consequence of confident (or precise) beliefs over certain outcomes. Here, we offer an alternative account of behavioral variability using Rényi divergences and their associated variational bounds. Rényi bounds are analogous to the variational free energy (or evidence lower bound) and can be derived under the same assumptions. Importantly, these bounds provide a formal way to establish behavioral differences through an α parameter, given fixed priors. This rests on changes in α that alter the bound (on a continuous scale),

inducing different posterior estimates and consequent variations in behavior. Thus, it looks as if individuals have different priors and have reached different conclusions. More specifically, $\alpha \to 0+$ optimization constrains the variational posterior to be positive whenever the true posterior is positive. This leads to mass-covering variational estimates and increased variability in choice behavior. Furthermore, $\alpha \to +\infty$ optimization constrains the variational posterior to be zero whenever the true posterior is zero. This leads to mass-seeking variational posteriors and greedy preferences. We exemplify this formulation through simulations of the multiarmed bandit task. We note that these α parameterizations may be especially relevant (i.e., shape preferences) when the true posterior is not in the same family of distributions as the assumed (simpler) approximate density, which may be the case in many real-world scenarios. The ensuing departure from vanilla variational inference provides a potentially useful explanation for differences in behavioral preferences of biological (or artificial) agents under the assumption that the brain performs variational Bayesian inference.

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25. Formulation and Emulation of Quantum-Inspired Dynamical Systems With Classical Analog Circuits

A. J. Cressman, Woradorn Wattanapanitch, Isaac L. Chuang, Rahul Sarpeshkar

https://doi.org/10.1162/neco_a_01481 | pp. 856-890

Quantum dynamical systems are capable of powerful computation but are hard to emulate on digital computers. We show that four novel analog circuit parts can emulate the phase-coherent unitary dynamics of such systems. These four parts are: a Planck capacitance analogous to a neuronal membrane capacitance; a quantum admittance element, together with the Planck capacitance, analogous to a neuronal quadrature oscillator; a quantum transadmittance element analogous to a complex neuronal synapse; and a quantum transadmittance mixer element analogous to a complex neuronal synapse with resonant modulation. These parts may be emulated classically, with paired real-value voltages on paired Planck capacitances corresponding to the real and imaginary portions of a probability amplitude; and appropriate paired real-value currents onto these Planck capacitances corresponding to diagonal (admittance), off-diagonal (transadmittance), or controlled off-diagonal (transadmittance mixer) Hamiltonian energy terms. The superposition of 2n simultaneously phase-coherent and symmetric probability-voltage amplitudes with O(n) of these parts, in a tensor-product architecture enables analog emulation of the quantum Fourier transform (QFT). Implementation of our circuits on an analog integrated circuit in a 0.18 μm process yield experimental results consistent with mathematical theory and computer simulations for emulations of NMR, Josephson junction, and QFT dynamics. Our results suggest that linear oscillatory neuronal networks with pairs of complex subthreshold/nonspiking sine and cosine neurons that are coupled together via complex synapses to other such complex neurons can architect quantum-inspired computation with classical analog circuits. Thus, an analog-circuit mapping between quantum and neural computation, both of which exploit analog computation for powerful operation, can enable future synergies between these fields.

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26. Neural Circuits for Dynamics-Based Segmentation of Time Series

Tiberiu Te■ileanu, Siavash Golkar, Samaneh Nasiri, Anirvan M. Sengupta, Dmitri B. Chklovskii

https://doi.org/10.1162/neco_a_01476 | pp. 891-938

The brain must extract behaviorally relevant latent variables from the signals streamed by the sensory organs. Such latent variables are often encoded in the dynamics that generated the signal rather than in the specific realization of the waveform. Therefore, one problem faced by the brain is to segment time series based on underlying dynamics. We present two algorithms for performing this segmentation task that are biologically plausible, which we define as acting in a streaming setting and all learning rules being local. One algorithm is model based and can be derived from an optimization problem involving a mixture of autoregressive processes. This algorithm relies on feedback in the form of a prediction error and can also be used for forecasting future samples. In some brain regions, such as the retina, the feedback connections necessary to use the prediction error for learning are absent. For this case, we propose a second, model-free algorithm that uses a running estimate of the autocorrelation structure of the signal to perform the segmentation. We show that both algorithms do well when tasked with segmenting signals drawn from autoregressive models with piecewise-constant parameters. In particular, the segmentation accuracy is similar to that obtained from oracle-like methods in which the ground-truth parameters of the autoregressive models are known. We also test our methods on data sets generated by alternating snippets of voice recordings. We provide implementations of our algorithms at https://github.com/ttesileanu/bio-time-series.

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27. Parameter Identification Problem in the Hodgkin-Huxley Model

Jemy A. Mandujano Valle, Alexandre L. Madureira

https://doi.org/10.1162/neco_a_01487 | pp. 939-970

The Hodgkin-Huxley (H-H) landmark model is described by a system of four nonlinear differential equations that describes how action potentials in neurons are initiated and propagated. However, obtaining some of the parameters of the model requires a tedious combination of experiments and data tuning. In this letter, we propose the use of a minimal error iteration method to estimate some of the parameters in the H-H model, given the measurements of membrane potential. We provide numerical results showing that the approach approximates well some of the model's parameters, using the measured voltage as data, even in the presence of noise.

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28. Adaptive Learning Neural Network Method for Solving Time–Fractional Diffusion Equations

Babak Shiri, Hua Kong, Guo–Cheng Wu, Cheng Luo

https://doi.org/10.1162/neco_a_01482 | pp. 971-990

A neural network method for solving fractional diffusion equations is presented in this letter. An adaptive gradient descent method is proposed to minimize energy functions. Due to the memory effects of the fractional calculus, the gradient of energy function becomes much more complicated, and we suggest a simplified method. Numerical examples with one-layer and two-layer neurons show the effectiveness of the method.

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29. Understanding Dynamics of Nonlinear Representation Learning and Its Application

Kenji Kawaguchi, Linjun Zhang, Zhun Deng
https://doi.org/10.1162/neco_a_01483 | pp. 991-1018

Representations of the world environment play a crucial role in artificial intelligence. It is often inefficient to conduct reasoning and inference directly in the space of raw sensory representations, such as pixel values of images. Representation learning allows us to automatically discover suitable representations from raw sensory data. For example, given raw sensory data, a deep neural network learns nonlinear representations at its hidden layers, which are subsequently used for classification (or regression) at its output layer. This happens implicitly during training through minimizing a supervised or unsupervised loss. In this letter, we study the dynamics of such implicit nonlinear representation learning. We identify a pair of a new assumption and a novel condition, called the on-model structure assumption and the data architecture alignment condition. Under the on-model structure assumption, the data architecture alignment condition is shown to be sufficient for the global convergence and necessary for global optimality. Moreover, our theory explains how and when increasing network size does and does not improve the training behaviors in the practical regime. Our results provide practical guidance for designing a model structure; for example, the on-model structure assumption can be used as a justification for using a particular model structure instead of others. As an application, we then derive a new training framework, which satisfies the data architecture alignment condition without assuming it by automatically modifying any given training algorithm dependent on data and architecture. Given a standard training algorithm, the framework running its modified version is empirically shown to maintain competitive (practical) test performances while providing global convergence guarantees for deep residual neural networks with convolutions, skip connections, and batch normalization with standard benchmark data sets, including MNIST, CIFAR-10, CIFAR-100, Semeion, KMNIST, and SVHN.

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30. Comparison of the Representational Power of Random Forests, Binary Decision Diagrams, and Neural Networks

So Kumano, Tatsuya Akutsu
https://doi.org/10.1162/neco_a_01486 | pp. 1019-1044

In this letter, we compare the representational power of random forests, binary decision diagrams (BDDs), and neural networks in terms of the number of nodes. We assume that an axis-aligned function on a single variable is assigned to each edge in random forests and BDDs, and the activation functions of neural networks are sigmoid, rectified linear unit, or similar functions. Based on existing studies, we show that for any random forest, there exists an equivalent depth-3 neural network with a linear number of nodes. We also show that for any BDD with balanced width, there exists an equivalent shallow depth neural network with a polynomial number of nodes. These results suggest that even shallow neural networks have the same or higher representation power than deep random forests and deep BDDs. We also show that in some cases, an exponential number of nodes are required to express a given random forest by a random forest with a much fewer number of trees, which suggests that many trees are required for random forests to represent some specific knowledge efficiently.

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31. Research on Imbalanced Data Classification Based on Classroom-Like Generative Adversarial Networks

Yancheng Lv, Lin Lin, Jie Liu, Hao Guo, Changsheng Tong
https://doi.org/10.1162/neco_a_01470 | pp. 1045-1073

Most of the research on machine learning classification methods is based on balanced data; the research on imbalanced data classification needs improvement. Generative adversarial networks (GANs) are able to learn high-dimensional complex data distribution without relying on a prior hypothesis, which has become a hot technology in artificial intelligence. In this letter, we propose a new structure, classroom-like generative adversarial networks (CLGANs), to construct a model with multiple generators. Taking inspiration from the fact that teachers arrange teaching activities according to students' learning situation, we propose a weight allocation function to adaptively adjust the influence weight of generator loss function on discriminator loss function. All the generators work together to improve the degree of discriminator and training sample space, so that a discriminator with excellent performance is trained and applied to the tasks of imbalanced data classification. Experimental results on the Case Western Reserve University data set and 2.4 GHz Indoor Channel Measurements data set show that the data classification ability of the discriminator trained by CLGANs with multiple generators is superior to that of other imbalanced data classification models, and the optimal discriminator can be obtained by selecting the right matching scheme of the generator models.

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32. Understanding the Computational Demands Underlying Visual Reasoning

Mohit Vaishnav, Rémi Cadène, Andrea Alamia, Drew Linsley, Rufin VanRullen, T. Serre
https://doi.org/10.1162/neco_a_01485 | pp. 1075-1099

Visual understanding requires comprehending complex visual relations between objects within a scene. Here, we seek to characterize the computational demands for abstract visual reasoning. We do this by systematically assessing the ability of modern deep convolutional neural networks (CNNs) to learn to solve the synthetic visual reasoning test (SVRT) challenge, a collection of 23 visual reasoning problems. Our analysis reveals a novel taxonomy of visual reasoning tasks, which can be primarily explained by both the type of relations (same-different versus spatial-relation judgments) and the number of relations used to compose the underlying rules. Prior cognitive neuroscience work suggests that attention plays a key role in humans' visual reasoning ability. To test this hypothesis, we extended the CNNs with spatial and feature-based attention mechanisms. In a second series of experiments, we evaluated the ability of these attention networks to learn to solve the SVRT challenge and found the resulting architectures to be much more efficient at solving the hardest of these visual reasoning tasks. Most important, the corresponding improvements on individual tasks partially explained our novel taxonomy. Overall, this work provides a granular computational account of visual reasoning and yields testable neuroscience predictions regarding the differential need for feature-based versus spatial attention depending on the type of visual reasoning problem.

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33. Direct Discriminative Decoder Models for Analysis of High-Dimensional Dynamical Neural Data

Mohammad R. Rezaei, Alex E. Hadjinicolaou, Sydney S. Cash, Uri T. Eden, Ali Yousefi
https://doi.org/10.1162/neco_a_01491 | pp. 1100-1135

With the accelerated development of neural recording technology over the past few decades, research in integrative neuroscience has become increasingly reliant on data analysis methods that are scalable to high-dimensional recordings and computationally tractable. Latent process models have shown promising results in estimating the dynamics of cognitive processes using individual models for each neuron's receptive field. However, scaling these models to work on high-dimensional neural recordings remains challenging. Not only is it impractical to build receptive field models for individual neurons of a large neural population, but most neural data analyses based on individual receptive field models discard the local history of neural activity, which has been shown to be critical in the accurate inference of the underlying cognitive processes. Here, we propose a novel, scalable latent process model that can directly estimate cognitive process dynamics without requiring precise receptive field models of individual neurons or brain nodes. We call this the direct discriminative decoder (DDD) model. The DDD model consists of (1) a discriminative process that characterizes the conditional distribution of the signal to be estimated, or state, as a function of both the current neural activity and its local history, and (2) a state transition model that characterizes the evolution of the state over a longer time period. While this modeling framework inherits advantages of existing latent process modeling methods, its computational cost is tractable. More important, the solution can incorporate any information from the history of neural activity at any timescale in computing the estimate of the state process. There are many choices in building the discriminative process, including deep neural networks or gaussian processes, which adds to the flexibility of the framework. We argue that these attributes of the proposed methodology, along with its applicability to different modalities of neural data, make it a powerful tool for high-dimensional neural data analysis. We also introduce an extension of these methods, called the discriminative-generative decoder (DGD). The DGD includes both discriminative and generative processes in characterizing observed data. As a result, we can combine physiological correlates like behavior with neural data to better estimate underlying cognitive processes. We illustrate the methods, including steps for inference and model identification, and demonstrate applications to multiple data analysis problems with high-dimensional neural recordings. The modeling results demonstrate the computational and modeling advantages of the DDD and DGD methods.

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34. On Neural Network Kernels and the Storage Capacity Problem

Jacob A. Zavatone-Veth, Cengiz Pehlevan

https://doi.org/10.1162/neco_a_01494 | pp. 1136-1142

In this short note, we reify the connection between work on the storage capacity problem in wide two-layer treelike neural networks and the rapidly growing body of literature on kernel limits of wide neural networks. Concretely, we observe that the "effective order parameter" studied in the statistical mechanics literature is exactly equivalent to the infinite-width neural network gaussian process kernel. This correspondence connects the expressivity and trainability of wide two-layer neural networks.

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35. Inference of Multiplicative Factors Underlying Neural Variability in Calcium Imaging Data

Marcus A. Triplett, Geoffrey J. Goodhill

https://doi.org/10.1162/neco_a_01492 | pp. 1143-1169

Understanding brain function requires disentangling the high-dimensional activity of populations of neurons. Calcium imaging is an increasingly popular technique for monitoring such neural activity, but computational tools for interpreting extracted calcium signals are lacking. While there has been a substantial development of factor analysis-type methods for neural spike train analysis, similar methods targeted at calcium imaging data are only beginning to emerge. Here we develop a flexible modeling framework that identifies low-dimensional latent factors in calcium imaging data with distinct additive and multiplicative modulatory effects. Our model includes spike-and-slab sparse priors that regularize additive factor activity and gaussian process priors that constrain multiplicative effects to vary only gradually, allowing for the identification of smooth and interpretable changes in multiplicative gain. These factors are estimated from the data using a variational expectation-maximization algorithm that requires a differentiable reparameterization of both continuous and discrete latent variables. After demonstrating our method on simulated data, we apply it to experimental data from the zebrafish optic tectum, uncovering low-dimensional fluctuations in multiplicative excitability that govern trial-to-trial variation in evoked responses.

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36. Training Deep Convolutional Spiking Neural Networks With Spike Probabilistic Global Pooling

Shuang Lian, Qianhui Liu, Rui Yan, Gang Pan, Huajin Tang

https://doi.org/10.1162/neco_a_01480 | pp. 1170-1188

Recent work on spiking neural networks (SNNs) has focused on achieving deep architectures. They commonly use backpropagation (BP) to train SNNs directly, which allows SNNs to go deeper and achieve higher performance. However, the BP training procedure is computing intensive and complicated by many trainable parameters. Inspired by global pooling in convolutional neural networks (CNNs), we present the spike probabilistic global pooling (SPGP) method based on a probability function for training deep convolutional SNNs. It aims to remove the difficulty of too many trainable parameters brought by multiple layers in the training process, which can reduce the risk of overfitting and get better performance for deep SNNs (DSNNs). We use the discrete leaky-integrate-fire model and the spatiotemporal BP algorithm for training DSNNs directly. As a result, our model trained with the SPGP method achieves competitive performance compared to the existing DSNNs on image and neuromorphic data sets while minimizing the number of trainable parameters. In addition, the proposed SPGP method shows its effectiveness in performance improvement, convergence, and generalization ability.

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37. Principal Component Analysis for Gaussian Process Posteriors

Hideaki Ishibashi, Shotaro Akaho

https://doi.org/10.1162/neco_a_01489 | pp. 1189-1219

This letter proposes an extension of principal component analysis for gaussian process (GP) posteriors, denoted by GP-PCA. Since GP-PCA estimates a low-dimensional space of GP posteriors, it can be used for metalearning, a framework for improving the performance of target tasks by estimating a structure of a set of tasks. The issue is how to define a structure of a set of GPs with an infinite-dimensional parameter, such as coordinate system and a divergence. In this study, we reduce the infiniteness of GP to the finite-dimensional case under the information geometrical framework by considering a space of GP posteriors that have the same prior. In addition, we propose an approximation method of GP-PCA based on variational inference and

demonstrate the effectiveness of GP-PCA as meta-learning through experiments.

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38. eSPA+: Scalable Entropy-Optimal Machine Learning Classification for Small Data Problems

Edoardo Vecchi, Lukáš Pospíšil, Steffen Albrecht, T. Okane, Illia Horenko

https://doi.org/10.1162/neco_a_01490 | pp. 1220-1255

Classification problems in the small data regime (with small data statistic T and relatively large feature space dimension D) impose challenges for the common machine learning (ML) and deep learning (DL) tools. The standard learning methods from these areas tend to show a lack of robustness when applied to data sets with significantly fewer data points than dimensions and quickly reach the overfitting bound, thus leading to poor performance beyond the training set. To tackle this issue, we propose eSPA+, a significant extension of the recently formulated entropy-optimal scalable probabilistic approximation algorithm (eSPA). Specifically, we propose to change the order of the optimization steps and replace the most computationally expensive subproblem of eSPA with its closed-form solution. We prove that with these two enhancements, eSPA+ moves from the polynomial to the linear class of complexity scaling algorithms. On several small data learning benchmarks, we show that the eSPA+ algorithm achieves a many-fold speed-up with respect to eSPA and even better performance results when compared to a wide array of ML and DL tools. In particular, we benchmark eSPA+ against the standard eSPA and the main classes of common learning algorithms in the small data regime: various forms of support vector machines, random forests, and long short-term memory algorithms. In all the considered applications, the common learning methods and eSPA are markedly outperformed by eSPA+, which achieves significantly higher prediction accuracy with an orders-of-magnitude lower computational cost.

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39. Graph Clustering With Graph Capsule Network

Xianchao Zhang, Jie Mu, Han Liu, Xiaotong Zhang, Linlin Zong, Guanglu Wang

https://doi.org/10.1162/neco_a_01493 | pp. 1256-1287

Graph clustering, which aims to partition a set of graphs into groups with similar structures, is a fundamental task in data analysis. With the great advances made by deep learning, deep graph clustering methods have achieved success. However, these methods have two limitations: (1) they learn graph embeddings by a neural language model that fails to effectively express graph properties, and (2) they treat embedding learning and clustering as two isolated processes, so the learned embeddings are unsuitable for the subsequent clustering. To overcome these limitations, we propose a novel capsule-based graph clustering (CGC) algorithm to cluster graphs. First, we construct a graph clustering capsule network (GCCN) that introduces capsules to capture graph properties. Second, we design an iterative optimization strategy to alternately update the GCCN parameters and clustering assignment parameters. This strategy leads GCCN to learn cluster-oriented graph embeddings. Experimental results show that our algorithm achieves performance superior to that of existing graph clustering algorithms in terms of three standard evaluation metrics: ACC, NMI, and ARI. Moreover, we use visualization results to analyze the effectiveness of the capsules and demonstrate that GCCN can learn cluster-oriented embeddings.

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40. Advancements in Algorithms and Neuromorphic Hardware for Spiking Neural Networks

Amirhossein Javanshir, Thanh Thi Nguyen, M. A. Parvez Mahmud, Abbas Z. Kouzani

https://doi.org/10.1162/neco_a_01499 | pp. 1289-1328

Artificial neural networks (ANNs) have experienced a rapid advancement for their success in various application domains, including autonomous driving and drone vision. Researchers have been improving the performance efficiency and computational requirement of ANNs inspired by the mechanisms of the biological brain. Spiking neural networks (SNNs) provide a power-efficient and brain-inspired computing paradigm for machine learning applications. However, evaluating large-scale SNNs on classical von Neumann architectures (central processing units/graphics processing units) demands a high amount of power and time. Therefore, hardware designers have developed neuromorphic platforms to execute SNNs in and approach that combines fast processing and low power consumption. Recently, field-programmable gate arrays (FPGAs) have been considered promising candidates for implementing neuromorphic solutions due to their varied advantages, such as higher flexibility, shorter design, and excellent stability. This review aims to describe recent advances in SNNs and the neuromorphic hardware platforms (digital, analog, hybrid, and FPGA based) suitable for their implementation. We present that biological background of SNN learning, such as neuron models and information encoding techniques, followed by a categorization of SNN training. In addition, we describe state-of-the-art SNN simulators. Furthermore, we review and present FPGA-based hardware implementation of SNNs. Finally, we discuss some future directions for research in this field.

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41. Predictive Coding Approximates Backprop Along Arbitrary Computation Graphs

Beren Millidge, Alexander Tschantz, Christopher L. Buckley

https://doi.org/10.1162/neco_a_01497 | pp. 1329-1368

Backpropagation of error (backprop) is a powerful algorithm for training machine learning architectures through end-to-end differentiation. Recently it has been shown that backprop in multilayer perceptrons (MLPs) can be approximated using predictive coding, a biologically plausible process theory of cortical computation that relies solely on local and Hebbian updates. The power of backprop, however, lies not in its instantiation in MLPs but in the concept of automatic differentiation, which allows for the optimization of any differentiable program expressed as a computation graph. Here, we demonstrate that predictive coding converges asymptotically (and in practice, rapidly) to exact backprop gradients on arbitrary computation graphs using only local learning rules. We apply this result to develop a straightforward strategy to translate core machine learning architectures into their predictive coding equivalents. We construct predictive coding convolutional neural networks, recurrent neural networks, and the more complex long short-term memory, which include a nonlayer-like branching internal graph structure and multiplicative interactions. Our models perform equivalently to backprop on challenging machine learning benchmarks while using only local and (mostly) Hebbian plasticity. Our method raises the potential that standard machine learning algorithms could in principle be directly implemented in neural circuitry and may also contribute to the development of completely distributed neuromorphic architectures.

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42. Decoding Pixel-Level Image Features From Two-Photon Calcium Signals of Macaque Visual Cortex

Yijun Zhang, Tong Bu, Jiyuan Zhang, Shiming Tang, Zhaofei Yu, Jian K. Liu, Tiejun Huang

https://doi.org/10.1162/neco_a_01498 | pp. 1369-1397

Images of visual scenes comprise essential features important for visual cognition of the brain. The complexity of visual features lies at different levels, from simple artificial patterns to natural images with different scenes. It has been a focus of using stimulus images to predict neural responses. However, it remains unclear how to extract features from neuronal responses. Here we address this question by leveraging two-photon calcium neural data recorded from the visual cortex of awake macaque monkeys. With stimuli including various categories of artificial patterns and diverse scenes of natural images, we employed a deep neural network decoder inspired by image segmentation technique. Consistent with the notation of sparse coding for natural images, a few neurons with stronger responses dominated the decoding performance, whereas decoding of ar tificial patterns needs a large number of neurons. When natural images using the model pretrained on artificial patterns are decoded, salient features of natural scenes can be extracted, as well as the conventional category information. Altogether, our results give a new perspective on studying neural encoding principles using reverse-engineering decoding strategies.

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43. Full-Span Log-Linear Model and Fast Learning Algorithm

Kazuya Takabatake, Shotaro Akaho

https://doi.org/10.1162/neco_a_01496 | pp. 1398-1424

The full-span log-linear (FSLL) model introduced in this letter is considered an nth order Boltzmann machine, where n is the number of all variables in the target system. Let X=(X0,…,Xn-1) be finite discrete random variables that can take |X|=|X0|…|Xn-1| different values. The FSLL model has |X|-1 parameters and can represent arbitrary positive distributions of X. The FSLL model is a highest-order Boltzmann machine; nevertheless, we can compute the dual parameter of the model distribution, which plays important roles in exponential families in O(|X|log|X|) time. Furthermore, using properties of the dual parameters of the FSLL model, we can construct an efficient learning algorithm. The FSLL model is limited to small probabilistic models up to |X|≈225; however, in this problem domain, the FSLL model flexibly fits various true distributions underlying the training data without any hyperparameter tuning. The experiments showed that the FSLL successfully learned six training data sets such that |X|=220 within 1 minute with a laptop PC.

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44. Role of Interaction Delays in the Synchronization of Inhibitory Networks

Nariman Roohi, Alireza Valizadeh

https://doi.org/10.1162/neco_a_01500 | pp. 1425-1447

Neural oscillations provide a means for efficient and flexible communication among different brain areas. Understanding the mechanisms of the generation of brain oscillations is crucial to determine principles of communication and information transfer in the brain circuits. It is well known that the inhibitory neurons play a major role in the generation of oscillations in the gamma range, in pure inhibitory networks, or in the networks composed of excitatory and inhibitory neurons. In this study, we explore the impact of different parameters and, in particular, the delay in the transmission of the signals between the neurons, on the

dynamics of inhibitory networks. We show that increasing delay in a reasonable range increases the synchrony and stabilizes the oscillations. Unstable gamma oscillations characterized by a highly variable amplitude of oscillations can be observed in an intermediate range of delays. We show that in this range of delays, other experimentally observed phenomena such as sparse firing, variable amplitude and period, and the correlation between the instantaneous amplitude and period could be observed. The results broaden our understanding of the mechanism of the generation of the gamma oscillations in the inhibitory networks, known as the ING (interneuron-gamma) mechanism.

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45. Hypothesis Test and Confidence Analysis With Wasserstein Distance on General Dimension

Masaaki Imaizumi, Hirofumi Ota, Takuo Hamaguchi

https://doi.org/10.1162/neco_a_01501 | pp. 1448-1487

We develop a general framework for statistical inference with the 1-Wasserstein distance. Recently, the Wasserstein distance has attracted considerable attention and has been widely applied to various machine learning tasks because of its excellent properties. However, hypothesis tests and a confidence analysis for it have not been established in a general multivariate setting. This is because the limit distribution of the empirical distribution with the Wasserstein distance is unavailable without strong restriction. To address this problem, in this study, we develop a novel nonasymptotic gaussian approximation for the empirical 1-Wasserstein distance. Using the approximation method, we develop a hypothesis test and confidence analysis for the empirical 1-Wasserstein distance. We also provide a theoretical guarantee and an efficient algorithm for the proposed approximation. Our experiments validate its performance numerically.

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46. The Perils of Being Unhinged: On the Accuracy of Classifiers Minimizing a Noise-Robust Convex Loss

Philip M. Long, Rocco A. Servedio

https://doi.org/10.1162/neco_a_01502 | pp. 1488-1499

van Rooyen, Menon, and Williamson (2015) introduced a notion of convex loss functions being robust to random classification noise and established that the "unhinged" loss function is robust in this sense. In this letter, we study the accuracy of binary classifiers obtained by minimizing the unhinged loss and observe that even for simple linearly separable data distributions, minimizing the unhinged loss may only yield a binary classifier with accuracy no better than random guessing.

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47. A Predictive Processing Model of Episodic Memory and Time Perception

Zafeirios Fountas, Anastasia Sylaidi, Kyriacos Nikiforou, Anil K. Seth, Murray Shanahan, Warrick Roseboom

https://doi.org/10.1162/neco_a_01514 | pp. 1501-1544

Human perception and experience of time are strongly influenced by ongoing stimulation, memory of past experiences, and required task context. When paying attention to time, time experience seems to expand; when distracted, it seems to contract. When considering time based on memory, the experience may be different than what is in the moment, exemplified by sayings like "time flies when you're having fun."

Experience of time also depends on the content of perceptual experience-rapidly changing or complex perceptual scenes seem longer in duration than less dynamic ones. The complexity of interactions among attention, memory, and perceptual stimulation is a likely reason that an overarching theory of time perception has been difficult to achieve. Here, we introduce a model of perceptual processing and episodic memory that makes use of hierarchical predictive coding, short-term plasticity, spatiotemporal attention, and episodic memory formation and recall, and apply this model to the problem of human time perception. In an experiment with approximately 13,000 human participants, we investigated the effects of memory, cognitive load, and stimulus content on duration reports of dynamic natural scenes up to about 1 minute long. Using our model to generate duration estimates, we compared human and model performance. Model-based estimates replicated key qualitative biases, including differences by cognitive load (attention), scene type (stimulation), and whether the judgment was made based on current or remembered experience (memory). Our work provides a comprehensive model of human time perception and a foundation for exploring the computational basis of episodic memory within a hierarchical predictive coding framework.

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48. Reduced-Dimension, Biophysical Neuron Models Constructed From Observed Data

Randall Clark, Lawson Fuller, Jason A. Platt, Henry D. I. Abarbanel

https://doi.org/10.1162/neco_a_01515 | pp. 1545-1587

Using methods from nonlinear dynamics and interpolation techniques from applied mathematics, we show how to use data alone to construct discrete time dynamical rules that forecast observed neuron properties. These data may come from simulations of a Hodgkin-Huxley (HH) neuron model or from laboratory current clamp experiments. In each case, the reduced-dimension, data-driven forecasting (DDF) models are shown to predict accurately for times after the training period. When the available observations for neuron preparations are, for example, membrane voltage V(t) only, we use the technique of time delay embedding from nonlinear dynamics to generate an appropriate space in which the full dynamics can be realized. The DDF constructions are reduced-dimension models relative to HH models as they are built on and forecast only observables such as V(t). They do not require detailed specification of ion channels, their gating variables, and the many parameters that accompany an HH model for laboratory measurements, yet all of this important information is encoded in the DDF model. As the DDF models use and forecast only voltage data, they can be used in building networks with biophysical connections. Both gap junction connections and ligand gated synaptic connections among neurons involve presynaptic voltages and induce postsynaptic voltage response. Biophysically based DDF neuron models can replace other reduced-dimension neuron models, say, of the integrate-and-fire type, in developing and analyzing large networks of neurons. When one does have detailed HH model neurons for network components, a reduced-dimension DDF realization of the HH voltage dynamics may be used in network computations to achieve computational efficiency and the exploration of larger biological networks.

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49. Differential Dopamine Receptor-Dependent Sensitivity Improves the Switch Between Hard and Soft Selection in a Model of the Basal Ganglia

Olivier Codol, Paul L. Gribble, Kevin Gurney

https://doi.org/10.1162/neco_a_01517 | pp. 1588-1615

The problem of selecting one action from a set of different possible actions, simply referred to as the problem of action selection, is a ubiquitous challenge in the animal world. For vertebrates, the basal ganglia (BG) are widely thought to implement the core computation to solve this problem, as its anatomy and physiology are well suited to this end. However, the BG still display physiological features whose role in achieving efficient action selection remains unclear. In particular, it is known that the two types of dopaminergic receptors (D1 and D2) present in the BG give rise to mechanistically different responses. The overall effect will be a difference in sensitivity to dopamine, which may have ramifications for action selection. However, which receptor type leads to a stronger response is unclear due to the complexity of the intracellular mechanisms involved. In this study, we use an existing, high-level computational model of the BG, which assumes that dopamine contributes to action selection by enabling a switch between different selection regimes, to predict which of D1 or D2 has the greater sensitivity. Thus, we ask, Assuming dopamine enables a switch between action selection regimes in the BG, what functional sensitivity values would result in improved action selection computation? To do this, we quantitatively assessed the model's capacity to perform action selection as we parametrically manipulated the sensitivity weights of D1 and D2. We show that differential (rather than equal) D1 and D2 sensitivity to dopaminergic input improves the switch between selection regimes during the action selection computation in our model. Specifically, greater D2 sensitivity compared to D1 led to these improvements.

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50. Sensitivity of Sparse Codes to Image Distortions

Kyle Luther, H. Sebastian Seung

https://doi.org/10.1162/neco_a_01513 | pp. 1616-1635

Sparse coding has been proposed as a theory of visual cortex and as an unsupervised algorithm for learning representations. We show empirically with the MNIST data set that sparse codes can be very sensitive to image distortions, a behavior that may hinder invariant object recognition. A locally linear analysis suggests that the sensitivity is due to the existence of linear combinations of active dictionary elements with high cancellation. A nearest-neighbor classifier is shown to perform worse on sparse codes than original images. For a linear classifier with a sufficiently large number of labeled examples, sparse codes are shown to yield higher accuracy than original images, but no higher than a representation computed by a random feedforward net. Sensitivity to distortions seems to be a basic property of sparse codes, and one should be aware of this property when applying sparse codes to invariant object recognition.

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51. Using Global t-SNE to Preserve Intercluster Data Structure

Yuansheng Zhou, Tatyana O. Sharpee

https://doi.org/10.1162/neco_a_01504 | pp. 1637-1651

The t-distributed stochastic neighbor embedding (t-SNE) method is one of the leading techniques for data visualization and clustering. This method finds lower-dimensional embedding of data points while minimizing distortions in distances between neighboring data points. By construction, t-SNE discards information about large-scale structure of the data. We show that adding a global cost function to the t-SNE cost function makes it possible to cluster the data while preserving global intercluster data structure. We test the new global t-SNE (g-SNE) method on one synthetic and two real data sets on flower shapes and human brain cells. We find that significant and meaningful global structure exists in both the plant and human brain data

sets. In all cases, g-SNE outperforms t-SNE and UMAP in preserving the global structure. Topological analysis of the clustering result makes it possible to find an appropriate trade-off of data distribution across scales. We find differences in how data are distributed across scales between the two subjects that were part of the human brain data set. Thus, by striving to produce both accurate clustering and positioning between clusters, the g-SNE method can identify new aspects of data organization across scales.

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52. Recurrent Connections in the Primate Ventral Visual Stream Mediate a Trade-Off Between Task Performance and Network Size During Core Object Recognition

Aran Nayebi, Javier Sagastuy-Breña, Daniel M. Bear, Kohitij Kar, Jonas Kubilius, Surya Ganguli, David Sussillo, James J. DiCarlo, Daniel Yamins

https://doi.org/10.1162/neco_a_01506 | pp. 1652-1675

The computational role of the abundant feedback connections in the ventral visual stream is unclear, enabling humans and nonhuman primates to effortlessly recognize objects across a multitude of viewing conditions. Prior studies have augmented feedforward convolutional neural networks (CNNs) with recurrent connections to study their role in visual processing; however, often these recurrent networks are optimized directly on neural data or the comparative metrics used are undefined for standard feedforward networks that lack these connections. In this work, we develop task-optimized convolutional recurrent (ConvRNN) network models that more correctly mimic the timing and gross neuroanatomy of the ventral pathway. Properly chosen intermediate-depth ConvRNN circuit architectures, which incorporate mechanisms of feedforward bypassing and recurrent gating, can achieve high performance on a core recognition task, comparable to that of much deeper feedforward networks. We then develop methods that allow us to compare both CNNs and ConvRNNs to finely grained measurements of primate categorization behavior and neural response trajectories across thousands of stimuli. We find that high-performing ConvRNNs provide a better match to these data than feedforward networks of any depth, predicting the precise timings at which each stimulus is behaviorally decoded from neural activation patterns. Moreover, these ConvRNN circuits consistently produce quantitatively accurate predictions of neural dynamics from V4 and IT across the entire stimulus presentation. In fact, we find that the highest-performing ConvRNNs, which best match neural and behavioral data, also achieve a strong Pareto trade-off between task performance and overall network size. Taken together, our results suggest the functional purpose of recurrence in the ventral pathway is to fit a high-performing network in cortex, attaining computational power through temporal rather than spatial complexity.

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53. Learning and Inference in Sparse Coding Models With Langevin Dynamics

Michael Y. -S. Fang, Mayur Mudigonda, Ryan Zarcone, Amir Khosrowshahi, Bruno A. Olshausen

https://doi.org/10.1162/neco_a_01505 | pp. 1676-1700

We describe a stochastic, dynamical system capable of inference and learning in a probabilistic latent variable model. The most challenging problem in such models-sampling the posterior distribution over latent variables-is proposed to be solved by harnessing natural sources of stochasticity inherent in electronic and neural systems. We demonstrate this idea for a sparse coding model by deriving a continuous-time equation

for inferring its latent variables via Langevin dynamics. The model parameters are learned by simultaneously evolving according to another continuous-time equation, thus bypassing the need for digital accumulators or a global clock. Moreover, we show that Langevin dynamics lead to an efficient procedure for sampling from the posterior distribution in the L0 sparse regime, where latent variables are encouraged to be set to zero as opposed to having a small L1 norm. This allows the model to properly incorporate the notion of sparsity rather than having to resort to a relaxed version of sparsity to make optimization tractable. Simulations of the proposed dynamical system on both synthetic and natural image data sets demonstrate that the model is capable of probabilistically correct inference, enabling learning of the dictionary as well as parameters of the prior.

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54. A Dynamic Neural Field Model of Multimodal Merging: Application to the Ventriloquist Effect

Simon Forest, Jean■Charles Quinton, Mathieu Lefort

https://doi.org/10.1162/neco_a_01509 | pp. 1701-1726

Multimodal merging encompasses the ability to localize stimuli based on imprecise information sampled through individual senses such as sight and hearing. Merging decisions are standardly described using Bayesian models that fit behaviors over many trials, encapsulated in a probability distribution. We introduce a novel computational model based on dynamic neural fields able to simulate decision dynamics and generate localization decisions, trial by trial, adapting to varying degrees of discrepancy between audio and visual stimulations. Neural fields are commonly used to model neural processes at a mesoscopic scale-for instance, neurophysiological activity in the superior colliculus. Our model is fit to human psychophysical data of the ventriloquist effect, additionally testing the influence of retinotopic projection onto the superior colliculus and providing a quantitative performance comparison to the Bayesian reference model. While models perform equally on average, a qualitative analysis of free parameters in our model allows insights into the dynamics of the decision and the individual variations in perception caused by noise. We finally show that the increase in the number of free parameters does not result in overfitting and that the parameter space may be either reduced to fit specific criteria or exploited to perform well on more demanding tasks in the future. Indeed, beyond decision or localization tasks, our model opens the door to the simulation of behavioral dynamics, as well as saccade generation driven by multimodal stimulation.

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55. Fixed-Time Stable Neurodynamic Flow to Sparse Signal Recovery via Nonconvex L1-β2-Norm

You Zhao, Xiaofeng Liao, Xing He

https://doi.org/10.1162/neco_a_01508 | pp. 1727-1755

This letter develops a novel fixed-time stable neurodynamic flow (FTSNF) implemented in a dynamical system for solving the nonconvex, nonsmooth model L1-β2, $\beta \in [0,1]$ to recover a sparse signal. FTSNF is composed of many neuron-like elements running in parallel. It is very efficient and has provable fixed-time convergence. First, a closed-form solution of the proximal operator to model L1-β2, $\beta \in [0,1]$ is presented based on the classic soft thresholding of the L1-norm. Next, the proposed FTSNF is proven to have a fixed-time convergence property without additional assumptions on the convexity and strong monotonicity of the objective functions. In addition, we show that FTSNF can be transformed into other proximal

neurodynamic flows that have exponential and finite-time convergence properties. The simulation results of sparse signal recovery verify the effectiveness and superiority of the proposed FTSNF.

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56. Invariance, Encodings, and Generalization: Learning Identity Effects With Neural Networks

S. Brugiapaglia, M. Liu, P. Tupper

https://doi.org/10.1162/neco_a_01510 | pp. 1756-1789

Often in language and other areas of cognition, whether two components of an object are identical or not determines if it is well formed. We call such constraints identity effects. When developing a system to learn well-formedness from examples, it is easy enough to build in an identity effect. But can identity effects be learned from the data without explicit guidance? We provide a framework in which we can rigorously prove that algorithms satisfying simple criteria cannot make the correct inference. We then show that a broad class of learning algorithms, including deep feedforward neural networks trained via gradient-based algorithms (such as stochastic gradient descent or the Adam method), satisfies our criteria, dependent on the encoding of inputs. In some broader circumstances, we are able to provide adversarial examples that the network necessarily classifies incorrectly. Finally, we demonstrate our theory with computational experiments in which we explore the effect of different input encodings on the ability of algorithms to generalize to novel inputs. This allows us to show similar effects to those predicted by theory for more realistic methods that violate some of the conditions of our theoretical results.

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57. Differential Geometry Methods for Constructing Manifold-Targeted Recurrent Neural Networks

Federico Claudi, Tiago Branco

https://doi.org/10.1162/neco_a_01511 | pp. 1790-1811

Neural computations can be framed as dynamical processes, whereby the structure of the dynamics within a neural network is a direct reflection of the computations that the network performs. A key step in generating mechanistic interpretations within this computation through dynamics framework is to establish the link among network connectivity, dynamics, and computation. This link is only partly understood. Recent work has focused on producing algorithms for engineering artificial recurrent neural networks (RNN) with dynamics targeted to a specific goal manifold. Some of these algorithms require only a set of vectors tangent to the target manifold to be computed and thus provide a general method that can be applied to a diverse set of problems. Nevertheless, computing such vectors for an arbitrary manifold in a high-dimensional state space remains highly challenging, which in practice limits the applicability of this approach. Here we demonstrate how topology and differential geometry can be leveraged to simplify this task by first computing tangent vectors on a low-dimensional topological manifold and then embedding these in state space. The simplicity of this procedure greatly facilitates the creation of manifold-targeted RNNs, as well as the process of designing task-solving, on-manifold dynamics. This new method should enable the application of network engineering-based approaches to a wide set of problems in neuroscience and machine learning. Our description of how fundamental concepts from differential geometry can be mapped onto different aspects of neural dynamics is a further demonstration of how the language of differential geometry can enrich the conceptual framework for describing neural dynamics and computation.

58. An FPGA Accelerator for High-Speed Moving Objects Detection and Tracking With a Spike Camera

Yaoyu Zhu, Yu Zhang, Xiaodong Xie, Tiejun Huang

https://doi.org/10.1162/neco_a_01507 | pp. 1812-1839

Ultra-high-speed object detection and tracking are crucial in fields such as fault detection and scientific observation. Existing solutions to this task have deficiencies in processing speeds. To deal with this difficulty, we propose a neural-inspired ultra-high-speed moving object filtering, detection, and tracking scheme, as well as a corresponding accelerator based on a high-speed spike camera. We parallelize the filtering module and divide the detection module to accelerate the algorithm and balance latency among modules for the benefit of the task-level pipeline. To be specific, a block-based parallel computation model is proposed to accelerate the filtering module, and the detection module is accelerated by a parallel connected component labeling algorithm modeling spike sparsity and spatial connectivity of moving objects with a searching tree. The hardware optimizations include processing the LIF layer with a group of multiplexers to reduce ADD operations and replacing expensive exponential operations with multiplications of preprocessed fixed-point values to increase processing speed and minimize resource consumption. We design an accelerator with the above techniques, achieving 19 times acceleration over the serial version after 25-way parallelization. A processing system for the accelerator is also implemented on the Xilinx ZCU-102 board to validate its functionality and performance. Our accelerator can process more than 20,000 spike images with 250 × 400 resolution per second with 1.618 W dynamic power consumption.

59. A Model of Semantic Completion in Generative Episodic Memory

Zahra Fayyaz, Aya Altamimi, Carina Zoellner, Nicole Klein, Oliver T. Wolf, Sen Cheng, Laurenz Wiskott

https://doi.org/10.1162/neco_a_01520 | pp. 1841-1870

Many studies have suggested that episodic memory is a generative process, but most computational models adopt a storage view. In this article, we present a model of the generative aspects of episodic memory. It is based on the central hypothesis that the hippocampus stores and retrieves selected aspects of an episode as a memory trace, which is necessarily incomplete. At recall, the neocortex reasonably fills in the missing parts based on general semantic information in a process we call semantic completion. The model combines two neural network architectures known from machine learning, the vector-quantized variational autoencoder (VQ-VAE) and the pixel convolutional neural network (PixelCNN). As episodes, we use images of digits and fashion items (MNIST) augmented by different backgrounds representing context. The model is able to complete missing parts of a memory trace in a semantically plausible way up to the point where it can generate plausible images from scratch, and it generalizes well to images not trained on. Compression as well as semantic completion contribute to a strong reduction in memory requirements and robustness to noise. Finally, we also model an episodic memory experiment and can reproduce that semantically congruent contexts are always recalled better than incongruent ones, high attention levels improve memory accuracy in both cases, and contexts that are not remembered correctly are more often remembered semantically congruently than completely wrong. This model contributes to a deeper understanding of the interplay between episodic memory and semantic information in the generative process of recalling the past.

60. Probing the Relationship Between Latent Linear Dynamical Systems and Low-Rank Recurrent Neural Network Models

Adrian Valente, Srdjan Ostojic, Jonathan W. Pillow

https://doi.org/10.1162/neco_a_01522 | pp. 1871-1892

A large body of work has suggested that neural populations exhibit low-dimensional dynamics during behavior. However, there are a variety of different approaches for modeling low-dimensional neural population activity. One approach involves latent linear dynamical system (LDS) models, in which population activity is described by a projection of low-dimensional latent variables with linear dynamics. A second approach involves low-rank recurrent neural networks (RNNs), in which population activity arises directly from a low-dimensional projection of past activity. Although these two modeling approaches have strong similarities, they arise in different contexts and tend to have different domains of application. Here we examine the precise relationship between latent LDS models and linear low-rank RNNs. When can one model class be converted to the other, and vice versa? We show that latent LDS models can only be converted to RNNs in specific limit cases, due to the non-Markovian property of latent LDS models. Conversely, we show that linear RNNs can be mapped onto LDS models, with latent dimensionality at most twice the rank of the RNN. A surprising consequence of our results is that a partially observed RNN is better represented by an LDS model than by an RNN consisting of only observed units.

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61. Bayesian Integration in a Spiking Neural System for Sensorimotor Control

Massimo Grillo, Alice Geminiani, Cristiano Alessandro, Egidio D'Angelo, Alessandra Pedrocchi, Claudia Casellato

https://doi.org/10.1162/neco_a_01525 | pp. 1893-1914

The brain continuously estimates the state of body and environment, with specific regions that are thought to act as Bayesian estimator, optimally integrating noisy and delayed sensory feedback with sensory predictions generated by the cerebellum. In control theory, Bayesian estimators are usually implemented using high-level representations. In this work, we designed a new spike-based computational model of a Bayesian estimator. The state estimator receives spiking activity from two neural populations encoding the sensory feedback and the cerebellar prediction, and it continuously computes the spike variability within each population as a reliability index of the signal these populations encode. The state estimator output encodes the current state estimate. We simulated a reaching task at different stages of cerebellar learning. The activity of the sensory feedback neurons encoded a noisy version of the trajectory after actual movement, with an almost constant intrapopulation spiking variability. Conversely, the activity of the cerebellar output neurons depended on the phase of the learning process. Before learning, they fired at their baseline not encoding any relevant information, and the variability was set to be higher than that of the sensory feedback (more reliable, albeit delayed). When learning was complete, their activity encoded the trajectory before the actual execution, providing an accurate sensory prediction; in this case, the variability was set to be lower than that of the sensory feedback. The state estimator model optimally integrated the neural activities of the afferent populations, so that the output state estimate was primarily driven by sensory feedback in prelearning and by the cerebellar prediction in postlearning. It was able to deal even with more complex scenarios, for example, by shifting the dominant source during the movement execution if information availability suddenly changed. The proposed tool will be a critical block within integrated spiking, brain-inspired control systems for simulations of sensorimotor tasks.

62. Scalability of Large Neural Network Simulations via Activity Tracking With Time Asynchrony and Procedural Connectivity

Cyrille Mascart, Gilles Scarella, Patricia Reynaud-Bouret, Alexandre Muzy

https://doi.org/10.1162/neco_a_01524 | pp. 1915-1943

We present a new algorithm to efficiently simulate random models of large neural networks satisfying the property of time asynchrony. The model parameters (average firing rate, number of neurons, synaptic connection probability, and postsynaptic duration) are of the order of magnitude of a small mammalian brain or of human brain areas. Through the use of activity tracking and procedural connectivity (dynamical regeneration of synapses), computational and memory complexities of this algorithm are proved to be theoretically linear with the number of neurons. These results are experimentally validated by sequential simulations of millions of neurons and billions of synapses running in a few minutes using a single thread of an equivalent desktop computer.

63. Information Geometrically Generalized Covariate Shift Adaptation

Masanari Kimura, Hideitsu Hino

https://doi.org/10.1162/neco_a_01526 | pp. 1944-1977

Many machine learning methods assume that the training and test data follow the same distribution. However, in the real world, this assumption is often violated. In particular, the marginal distribution of the data changes, called covariate shift, is one of the most important research topics in machine learning. We show that the well-known family of covariate shift adaptation methods is unified in the framework of information geometry. Furthermore, we show that parameter search for a geometrically generalized covariate shift adaptation method can be achieved efficiently. Numerical experiments show that our generalization can achieve better performance than the existing methods it encompasses.

64. Permitted Sets and Convex Coding in Nonthreshold Linear Networks

Steven Collazos, Duane Q. Nykamp

https://doi.org/10.1162/neco_a_01523 | pp. 1978-2008

Hebbian theory proposes that ensembles of neurons form a basis for neural processing. It is possible to gain insight into the activity patterns of these neural ensembles through a binary analysis, regarding neurons as either active or inactive. The framework of permitted and forbidden sets, introduced by Hahnloser, Seung, and Slotine (2003), is a mathematical model of such a binary analysis: groups of coactive neurons can be permitted or forbidden depending on the network's structure. In order to widen the applicability of the framework of permitted sets, we extend the permitted set analysis from the original threshold-linear regime. Specifically, we generalize permitted sets to firing rate models in which Φ is a nonnegative continuous piecewise C1 activation function. In our framework, the focus is shifted from a neuron's firing rate to its responsiveness to inputs; if a neuron's firing rate is sufficiently sensitive to changes in its input, we say that the neuron is responsive. The algorithm for categorizing a neuron as responsive depends on thresholds that a user can select arbitrarily and that are independent of the dynamics. Given a synaptic weight matrix W, we say that a set of neurons is permitted if it is possible to find a stimulus where those neurons, and no others,

remain responsive. The main coding property we establish about PΦ(W), the collection of all permitted sets of the network, is that PΦ(W) is a convex code when W is almost rank one. This means that PΦ(W) in the low-rank regime can be realized as a neural code resulting from the pattern of overlaps of receptive fields that are convex.

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65. Disentangled Representation Learning and Generation With Manifold Optimization

Arun Pandey, Michaël Fanuel, Joachim Schreurs, Johan A. K. Suykens

https://doi.org/10.1162/neco_a_01528 | pp. 2009-2036

Disentanglement is a useful property in representation learning, which increases the interpretability of generative models such as variational autoencoders (VAE), generative adversarial models, and their many variants. Typically in such models, an increase in disentanglement performance is traded off with generation quality. In the context of latent space models, this work presents a representation learning framework that explicitly promotes disentanglement by encouraging orthogonal directions of variations. The proposed objective is the sum of an autoencoder error term along with a principal component analysis reconstruction error in the feature space. This has an interpretation of a restricted kernel machine with the eigenvector matrix valued on the Stiefel manifold. Our analysis shows that such a construction promotes disentanglement by matching the principal directions in the latent space with the directions of orthogonal variation in data space. In an alternating minimization scheme, we use the Cayley ADAM algorithm, a stochastic optimization method on the Stiefel manifold along with the Adam optimizer. Our theoretical discussion and various experiments show that the proposed model is an improvement over many VAE variants in terms of both generation quality and disentangled representation learning.

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66. On Suspicious Coincidences and Pointwise Mutual Information

Christopher K. I. Williams

https://doi.org/10.1162/neco_a_01533 | pp. 2037-2046

Barlow (1985) hypothesized that the co-occurrence of two events A and B is "suspicious" if P(A,B)■P(A)P(B). We first review classical measures of association for 2 × 2 contingency tables, including Yule's Y (Yule, 1912), which depends only on the odds ratio λ and is independent of the marginal probabilities of the table. We then discuss the mutual information (MI) and pointwise mutual information (PMI), which depend on the ratio P(A,B)/P(A)P(B), as measures of association. We show that once the effect of the marginals is removed, MI and PMI behave similarly to Y as functions of λ. The pointwise mutual information is used extensively in some research communities for flagging suspicious coincidences. We discuss the pros and cons of using it in this way, bearing in mind the sensitivity of the PMI to the marginals, with increased scores for sparser events.

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67. Astrocytes Learn to Detect and Signal Deviations From Critical Brain Dynamics

Vladimir A. Ivanov, Konstantinos P. Michmizos

https://doi.org/10.1162/neco_a_01532 | pp. 2047-2074

Astrocytes are nonneuronal brain cells that were recently shown to actively communicate with neurons and are implicated in memory, learning, and regulation of cognitive states. Interestingly, these information processing functions are also closely linked to the brain's ability to self-organize at a critical phase transition. Investigating the mechanistic link between astrocytes and critical brain dynamics remains beyond the reach of cellular experiments, but it becomes increasingly approachable through computational studies. We developed a biologically plausible computational model of astrocytes to analyze how astrocyte calcium waves can respond to changes in underlying network dynamics. Our results suggest that astrocytes detect synaptic activity and signal directional changes in neuronal network dynamics using the frequency of their calcium waves. We show that this function may be facilitated by receptor scaling plasticity by enabling astrocytes to learn the approximate information content of input synaptic activity. This resulted in a computationally simple, information-theoretic model, which we demonstrate replicating the signaling functionality of the biophysical astrocyte model with receptor scaling. Our findings provide several experimentally testable hypotheses that offer insight into the regulatory role of astrocytes in brain information processing.

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68. A Simple Model of Nonspiking Neurons

Loïs Naudin, Juan Luís Jiménez Laredo, Nathalie Corson

https://doi.org/10.1162/neco_a_01531 | pp. 2075-2101

Due to the ubiquity of spiking neurons in neuronal processes, various simple spiking neuron models have been proposed as an alternative to conductance-based models (a.k.a. Hodgkin-Huxley-type models), known to be computationally expensive and difficult to treat mathematically. However, to the best of our knowledge, there is no equivalent in the literature of a simple and lightweight model for describing the voltage behavior of nonspiking neurons, which also are ubiquitous in a large variety of nervous tissues in both vertebrate and invertebrate species and play a central role in information processing. This letter proposes a simple model that reproduces the experimental qualitative behavior of known types of nonspiking neurons. The proposed model, which differs fundamentally from classic simple spiking models unable to characterize nonspiking dynamics due to their intrinsic structure, is derived from the bifurcation study of conductance-based models of nonspiking neurons. Since such neurons display a high sensitivity to noise, the model aims at capturing the experimental distribution of single-neuron responses rather than perfectly replicating a single given experimental voltage trace. We show that such a model can be used as a building block for realistic simulations of large nonspiking neuronal networks and is endowed with generalization capabilities, granted by design.

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69. Neural Information Processing and Computations of Two-Input Synapses

Soon Ho Kim, Junhyuk Woo, Kiri Choi, M. Y. Choi, Kyungreem Han

https://doi.org/10.1162/neco_a_01534 | pp. 2102-2131

Information processing in artificial neural networks is largely dependent on the nature of neuron models. While commonly used models are designed for linear integration of synaptic inputs, accumulating experimental evidence suggests that biological neurons are capable of nonlinear computations for many converging synaptic inputs via homo- and heterosynaptic mechanisms. This nonlinear neuronal computation

may play an important role in complex information processing at the neural circuit level. Here we characterize the dynamics and coding properties of neuron models on synaptic transmissions delivered from two hidden states. The neuronal information processing is influenced by the cooperative and competitive interactions among synapses and the coherence of the hidden states. Furthermore, we demonstrate that neuronal information processing under two-input synaptic transmission can be mapped to linearly nonseparable XOR as well as basic AND/OR operations. In particular, the mixtures of linear and nonlinear neuron models outperform the fashion-MNIST test compared to the neural networks consisting of only one type. This study provides a computational framework for assessing information processing of neuron and synapse models that may be beneficial for the design of brain-inspired artificial intelligence algorithms and neuromorphic systems.

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70. Branching Time Active Inference with Bayesian Filtering

Théophile Champion, Marek Grze■, Howard Bowman

https://doi.org/10.1162/neco_a_01529 | pp. 2132-2144

Branching time active inference is a framework proposing to look at planning as a form of Bayesian model expansion. Its root can be found in active inference, a neuroscientific framework widely used for brain modeling, as well as in Monte Carlo tree search, a method broadly applied in the reinforcement learning literature. Up to now, the inference of the latent variables was carried out by taking advantage of the flexibility offered by variational message passing, an iterative process that can be understood as sending messages along the edges of a factor graph. In this letter, we harness the efficiency of an alternative method for inference, Bayesian filtering, which does not require the iteration of the update equations until convergence of the variational free energy. Instead, this scheme alternates between two phases: integration of evidence and prediction of future states. Both phases can be performed efficiently, and this provides a forty times speedup over the state of the art.

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71. A Generalized Framework of Multifidelity Max-Value Entropy Search Through Joint Entropy

Shion Takeno, Hitoshi Fukuoka, Yuhki Tsukada, Toshiyuki Koyama, M. Shiga, Ichiro Takeuchi, Masayuki Karasuyama

https://doi.org/10.1162/neco_a_01530 | pp. 2145-2203

Bayesian optimization (BO) is a popular method for expensive black-box optimization problems; however, querying the objective function at every iteration can be a bottleneck that hinders efficient search capabilities. In this regard, multifidelity Bayesian optimization (MFBO) aims to accelerate BO by incorporating lower-fidelity observations available with a lower sampling cost. In our previous work, we proposed an information-theoretic approach to MFBO, referred to as multifidelity max-value entropy search (MF-MES), which inherits practical effectiveness and computational simplicity of the well-known max-value entropy search (MES) for the single-fidelity BO. However, the applicability of MF-MES is still limited to the case that a single observation is sequentially obtained. In this letter, we generalize MF-MES so that information gain can be evaluated even when multiple observations are simultaneously obtained. This generalization enables MF-MES to address two practical problem settings: synchronous parallelization and trace-aware querying. We show that the acquisition functions for these extensions inherit the simplicity of

MF-MES without introducing additional assumptions. We also provide computational techniques for entropy evaluation and posterior sampling in the acquisition functions, which can be commonly used for all variants of MF-MES. The effectiveness of MF-MES is demonstrated using benchmark functions and real-world applications such as materials science data and hyperparameter tuning of machine-learning algorithms.

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72. A Neural Model for Insect Steering Applied to Olfaction and Path Integration

Andrea Adden, Terrence C. Stewart, Barbara Webb, Stanley Heinze

https://doi.org/10.1162/neco_a_01540 | pp. 2205-2231

Many animal behaviors require orientation and steering with respect to the environment. For insects, a key brain area involved in spatial orientation and navigation is the central complex. Activity in this neural circuit has been shown to track the insect's current heading relative to its environment and has also been proposed to be the substrate of path integration. However, it remains unclear how the output of the central complex is integrated into motor commands. Central complex output neurons project to the lateral accessory lobes (LAL), from which descending neurons project to thoracic motor centers. Here, we present a computational model of a simple neural network that has been described anatomically and physiologically in the LALs of male silkworm moths, in the context of odor-mediated steering. We present and analyze two versions of this network, one rate based and one based on spiking neurons. The modeled network consists of an inhibitory local interneuron and a bistable descending neuron (flip-flop) that both receive input in the LAL. The flip-flop neuron projects onto neck motor neurons to induce steering. We show that this simple computational model not only replicates the basic parameters of male silkworm moth behavior in a simulated odor plume but can also take input from a computational model of path integration in the central complex and use it to steer back to a point of origin. Furthermore, we find that increasing the level of detail within the model improves the realism of the model's behavior, leading to the emergence of looping behavior as an orientation strategy. Our results suggest that descending neurons originating in the LALs, such as flip-flop neurons, are sufficient to mediate multiple steering behaviors. This study is therefore a first step to close the gap between orientation circuits in the central complex and downstream motor centers.

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73. Recurrent Neural-Linear Posterior Sampling for Nonstationary Contextual Bandits

Aditya Ramesh, Paulo Rauber, Michelangelo Conserva, Jürgen Schmidhuber

https://doi.org/10.1162/neco_a_01539 | pp. 2232-2272

An agent in a nonstationary contextual bandit problem should balance between exploration and the exploitation of (periodic or structured) patterns present in its previous experiences. Handcrafting an appropriate historical context is an attractive alternative to transform a nonstationary problem into a stationary problem that can be solved efficiently. However, even a carefully designed historical context may introduce spurious relationships or lack a convenient representation of crucial information. In order to address these issues, we propose an approach that learns to represent the relevant context for a decision based solely on the raw history of interactions between the agent and the environment. This approach relies on a combination of features extracted by recurrent neural networks with a contextual linear bandit algorithm based on posterior sampling. Our experiments on a diverse selection of contextual and noncontextual

nonstationary problems show that our recurrent approach consistently outperforms its feedforward counterpart, which requires handcrafted historical contexts, while being more widely applicable than conventional nonstationary bandit algorithms. Although it is very difficult to provide theoretical performance guarantees for our new approach, we also prove a novel regret bound for linear posterior sampling with measurement error that may serve as a foundation for future theoretical work.

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74. Inferring Mechanisms of Auditory Attentional Modulation with Deep Neural Networks

Ting■Yu Kuo, Yuan-Da Liao, Kai Li, Bo Hong, Xiaolin Hu

https://doi.org/10.1162/neco_a_01537 | pp. 2273-2293

Humans have an exceptional ability to extract specific audio streams of interest in a noisy environment; this is known as the cocktail party effect. It is widely accepted that this ability is related to selective attention, a mental process that enables individuals to focus on a particular object. Evidence suggests that sensory neurons can be modulated by top-down signals transmitted from the prefrontal cortex. However, exactly how the projection of attention signals to the cortex and subcortex influences the cocktail effect is unclear. We constructed computational models to study whether attentional modulation is more effective at earlier or later stages for solving the cocktail party problem along the auditory pathway. We modeled the auditory pathway using deep neural networks (DNNs), which can generate representational neural patterns that resemble the human brain. We constructed a series of DNN models in which the main structures were autoencoders. We then trained these DNNs on a speech separation task derived from the dichotic listening paradigm, a common paradigm to investigate the cocktail party effect. We next analyzed the modulation effects of attention signals during all stages. Our results showed that the attentional modulation effect is more effective at the lower stages of the DNNs. This suggests that the projection of attention signals to lower stages within the auditory pathway plays a more significant role than the higher stages in solving the cocktail party problem. This prediction could be tested using neurophysiological experiments.

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75. Capacity Limitations of Visual Search in Deep Convolutional Neural Networks

Endel Põder

https://doi.org/10.1162/neco_a_01538 | pp. 2294-2308

Deep convolutional neural networks (CNN) follow roughly the architecture of biological visual systems and have shown a performance comparable to human observers in object classification tasks. In this study, three deep neural networks pretrained for image classification were tested in visual search for simple features and for feature configurations. The results reveal a qualitative difference from human performance. It appears that there is no clear difference between searches for simple features that pop out in experiments with humans and for feature configurations that exhibit strict capacity limitations in human vision. Both types of stimuli reveal comparable capacity limitations in the neural networks tested here.

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76. Beyond Backpropagation: Bilevel Optimization Through Implicit Differentiation and Equilibrium Propagation

Nicolas Zucchet, João Sacramento

https://doi.org/10.1162/neco_a_01547 | pp. 2309-2346

This review examines gradient-based techniques to solve bilevel optimization problems. Bilevel optimization extends the loss minimization framework underlying statistical learning to systems that are implicitly defined through a quantity they minimize. This characterization can be applied to neural networks, optimizers, algorithmic solvers, and even physical systems and allows for greater modeling flexibility compared to the usual explicit definition of such systems. We focus on solving learning problems of this kind through gradient descent, leveraging the toolbox of implicit differentiation and, for the first time applied to this setting, the equilibrium propagation theorem. We present the mathematical foundations behind such methods, introduce the gradient estimation algorithms in detail, and compare the competitive advantages of the different approaches.

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77. Large-Scale Algorithmic Search Identifies Stiff and Sloppy Dimensions in Synaptic Architectures Consistent With Murine Neocortical Wiring

Tarek Jabri, Jason N. MacLean

https://doi.org/10.1162/neco_a_01544 | pp. 2347-2373

Complex systems can be defined by "sloppy" dimensions, meaning that their behavior is unmodified by large changes to specific parameter combinations, and "stiff" dimensions, whose change results in considerable behavioral modification. In the neocortex, sloppiness in synaptic architectures would be crucial to allow for the maintenance of asynchronous irregular spiking dynamics with low firing rates despite a diversity of inputs, states, and short- and long-term plasticity. Using simulations on neural networks with first-order spiking statistics matched to firing in murine visual cortex while varying connectivity parameters, we determined the stiff and sloppy parameters of synaptic architectures across three classes of input (brief, continuous, and cyclical). Algorithmically generated connectivity parameter values drawn from a large portion of the parameter space reveal that specific combinations of excitatory and inhibitory connectivity are stiff and that all other architectural details are sloppy. Stiff dimensions are consistent across input classes with self-sustaining synaptic architectures following brief input occupying a smaller subspace as compared to the other input classes. Experimentally estimated connectivity probabilities from mouse visual cortex are consistent with the connectivity correlations found and fall in the same region of the parameter space as architectures identified algorithmically. This suggests that simple statistical descriptions of spiking dynamics are a sufficient and parsimonious description of neocortical activity when examining structure-function relationships at the mesoscopic scale. Additionally, coarse graining cell types does not prevent the generation of accurate, informative, and interpretable models underlying simple spiking activity. This unbiased investigation provides further evidence of the importance of the interrelationship of excitatory and inhibitory connectivity to establish and maintain stable spiking dynamical regimes in the neocortex.

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78. Memoryless Optimality: Neurons Do Not Need Adaptation to Optimally Encode Stimuli With Arbitrarily Complex Statistics

Oren Forkosh
https://doi.org/10.1162/neco_a_01543 | pp. 2374-2387

Our neurons seem capable of handling any type of data, regardless of its scale or statistical properties. In this letter, we suggest that optimal coding may occur at the single-neuron level without requiring memory, adaptation, or evolutionary-driven fit to the stimuli. We refer to a neural circuit as optimal if it maximizes the mutual information between its inputs and outputs. We show that often encountered differentiator neurons, or neurons that respond mainly to changes in the input, are capable of using all their information capacity when handling samples of any statistical distribution. We demonstrate this optimality using both analytical methods and simulations. In addition to demonstrating the simplicity and elegance of neural processing, this result might provide a way to improve the handling of data by artificial neural networks.

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79. Asymmetric Complexity in a Pupil Control Model With Laterally Imbalanced Neural Activity in the Locus Coeruleus: A Potential Biomarker for Attention-Deficit/Hyperactivity Disorder

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https://doi.org/10.1162/neco_a_01545 | pp. 2388-2407

Locus coeruleus (LC) overactivity, especially in the right hemisphere, is a recognized pathophysiology of attention-deficit/hyperactivity disorder (ADHD) and may be related to inattention. LC activity synchronizes with the kinetics of the pupil diameter and reflects neural activity related to cognitive functions such as attention and arousal. Recent studies highlight the importance of the complexity of the temporal patterns of pupil diameter. Moreover, asymmetrical pupil diameter, which correlates with the severity of inattention, impulsivity, and hyperactivity in ADHD, might be attributed to a left-right imbalance in LC activity. We recently constructed a computational model of pupil diameter based on the newly discovered contralateral projection from the LC to the Edinger-Westphal nucleus (EWN), which demonstrated mechanisms for the complex temporal patterns of pupil kinetics; however, it remains unclear how LC overactivity and its asymmetry affect pupil diameter. We hypothesized that a neural model of pupil diameter control featuring left-right differences in LC activity and projections onto two opponent sides may clarify the role of pupil behavior in ADHD studies. Therefore, we developed a pupil diameter control model reflecting LC overactivity in the right hemisphere by incorporating a contralateral projection from the LC to EWN and evaluated the complexity of the temporal patterns of pupil diameter generated by the model. Upon comparisons with experimentally measured pupil diameters in adult patients with ADHD, the parameter region of interest of the neural model was estimated, which was a region in the two-dimensional plot of complexity versus left-side LC baseline activity and that of the right. A region resulting in relatively high right-side complexity, which corresponded to the pathophysiological indexes, was identified. We anticipate that the discovery of lateralization of complexity in pupil diameter fluctuations will facilitate the development of biomarkers for accurate diagnosis of ADHD.

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80. Bayesian Optimization for Cascade-Type Multistage Processes

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https://doi.org/10.1162/neco_a_01550 | pp. 2408-2431

Complex processes in science and engineering are often formulated as multistage decision-making problems. In this letter, we consider a cascade process, a type of multistage decision-making process. This is a multistage process in which the output of one stage is used as an input for the subsequent stage. When the cost of each stage is expensive, it is difficult to search for the optimal controllable parameters for each stage exhaustively. To address this problem, we formulate the optimization of the cascade process as an extension of the Bayesian optimization framework and propose two types of acquisition functions based on credible intervals and expected improvement. We investigate the theoretical properties of the proposed acquisition functions and demonstrate their effectiveness through numerical experiments. In addition, we consider suspension setting, an extension in which we are allowed to suspend the cascade process at the middle of the multistage decision-making process that often arises in practical problems. We apply the proposed method in a test problem involving a solar cell simulator, the motivation for this study.

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81. Unsupervised Domain Adaptation for Extra Features in the Target Domain Using Optimal Transport

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https://doi.org/10.1162/neco_a_01549 | pp. 2432-2466

Domain adaptation aims to transfer knowledge of labeled instances obtained from a source domain to a target domain to fill the gap between the domains. Most domain adaptation methods assume that the source and target domains have the same dimensionality. Methods that are applicable when the number of features is different in each domain have rarely been studied, especially when no label information is given for the test data obtained from the target domain. In this letter, it is assumed that common features exist in both domains and that extra (new additional) features are observed in the target domain; hence, the dimensionality of the target domain is higher than that of the source domain. To leverage the homogeneity of the common features, the adaptation between these source and target domains is formulated as an optimal transport (OT) problem. In addition, a learning bound in the target domain for the proposed OT-based method is derived. The proposed algorithm is validated using both simulated and real-world data.